NEURO 208, School of Medicine, University of Crete

Lecture 4: Biophysical Models

Spyridon Chavlis, PhD

Athanasia Papoutsi, PhD

IMBB-FORTH

Recap

Simplified models

- Rate based models
- Integrate-and-fire models
- ~~Dynamical Systems analysis~~

Synapses

Biophysical Models

Compartmental Models

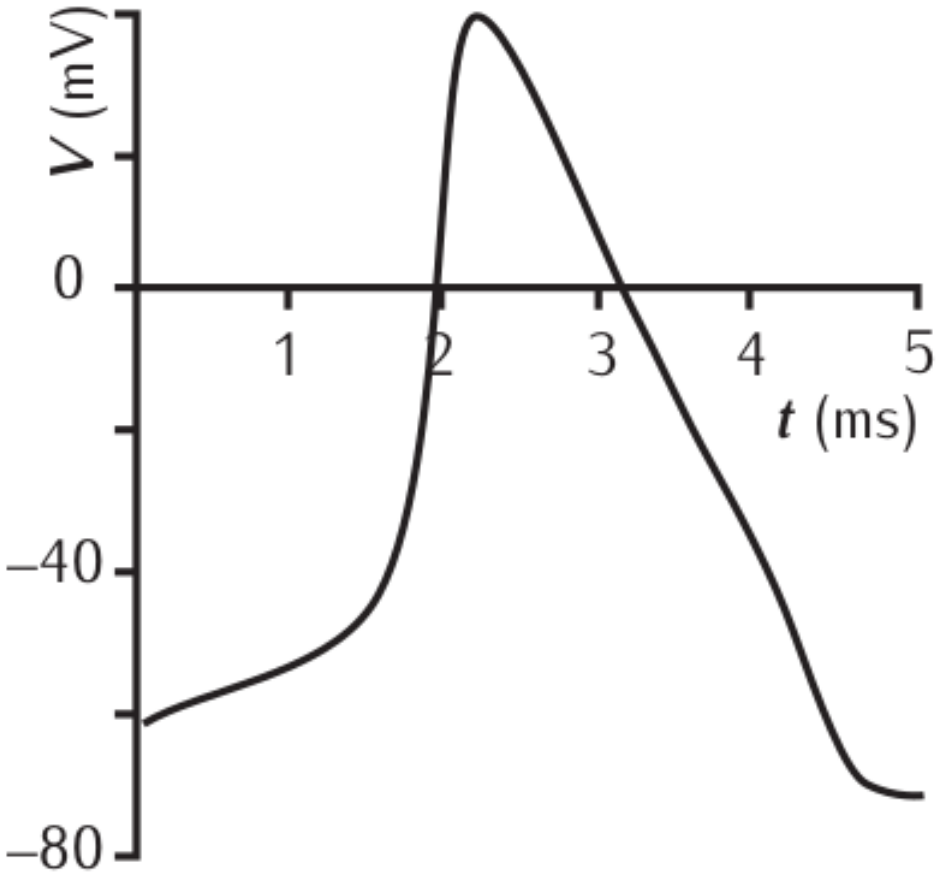
Action Potential

Sharp increase: **depolarization**

Less sharp decrease: **repolarization**

Afterhyperpolarization phase

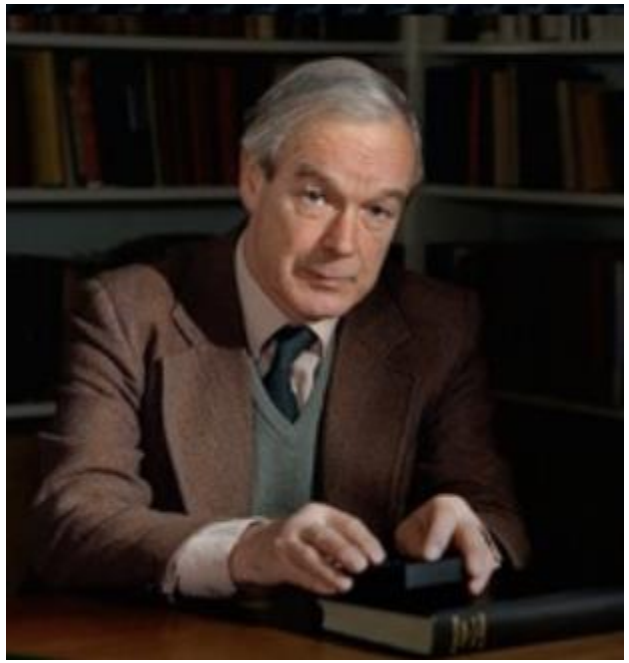
*Their magnitude does not decay during propagation (**regeneration**)*



Hodgkin-Huxley

- Hodgkin and Huxley (in collaboration with Katz) describe the **active** mechanisms quantitatively (Hodgkin et al, 1952, Hodgkin and Huxley, 1952a, b, c, d)
- They record **intracellularly** from the squid giant axon
- They fitted their results in a mathematical model (**HH model**)
- Properties of the simulated action potentials matched those of the recorded action potentials

Hodgkin-Huxley

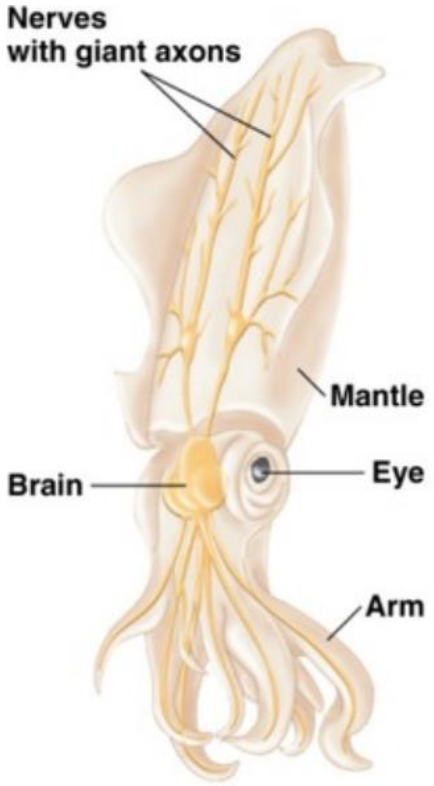


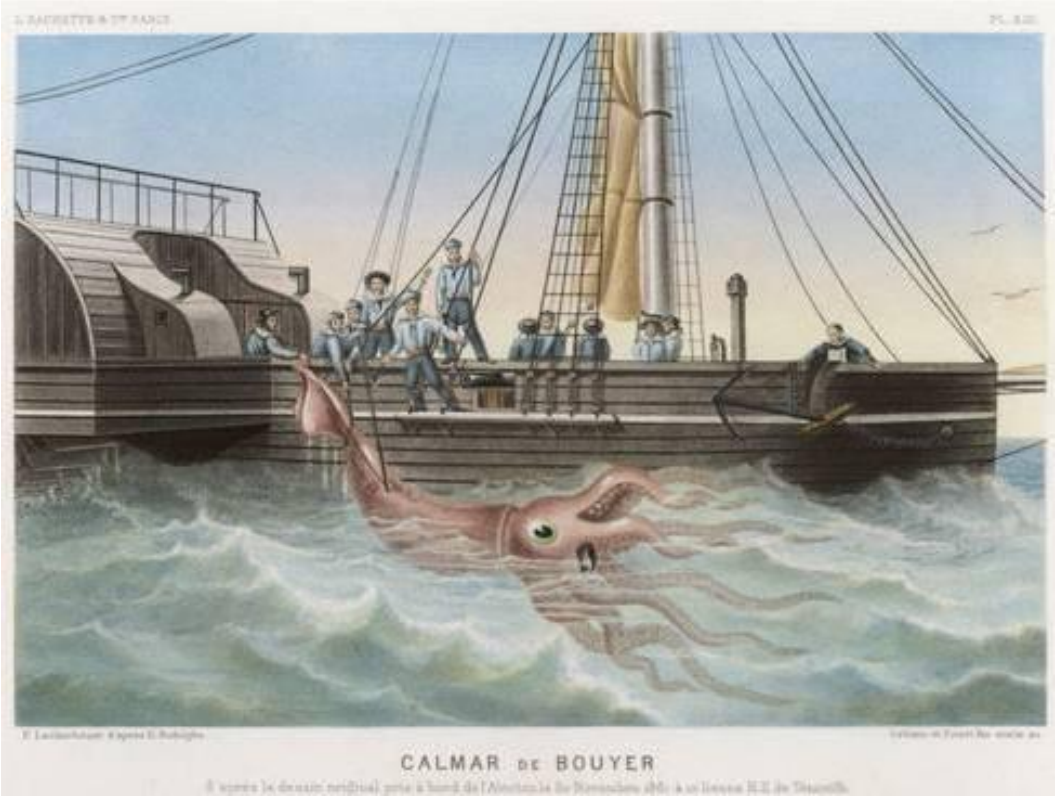
Nobel Prize in Medicine or Physiology in 1963

Work reflects a combination of experimental work, theoretical hypothesis, computational data fitting, and model prediction

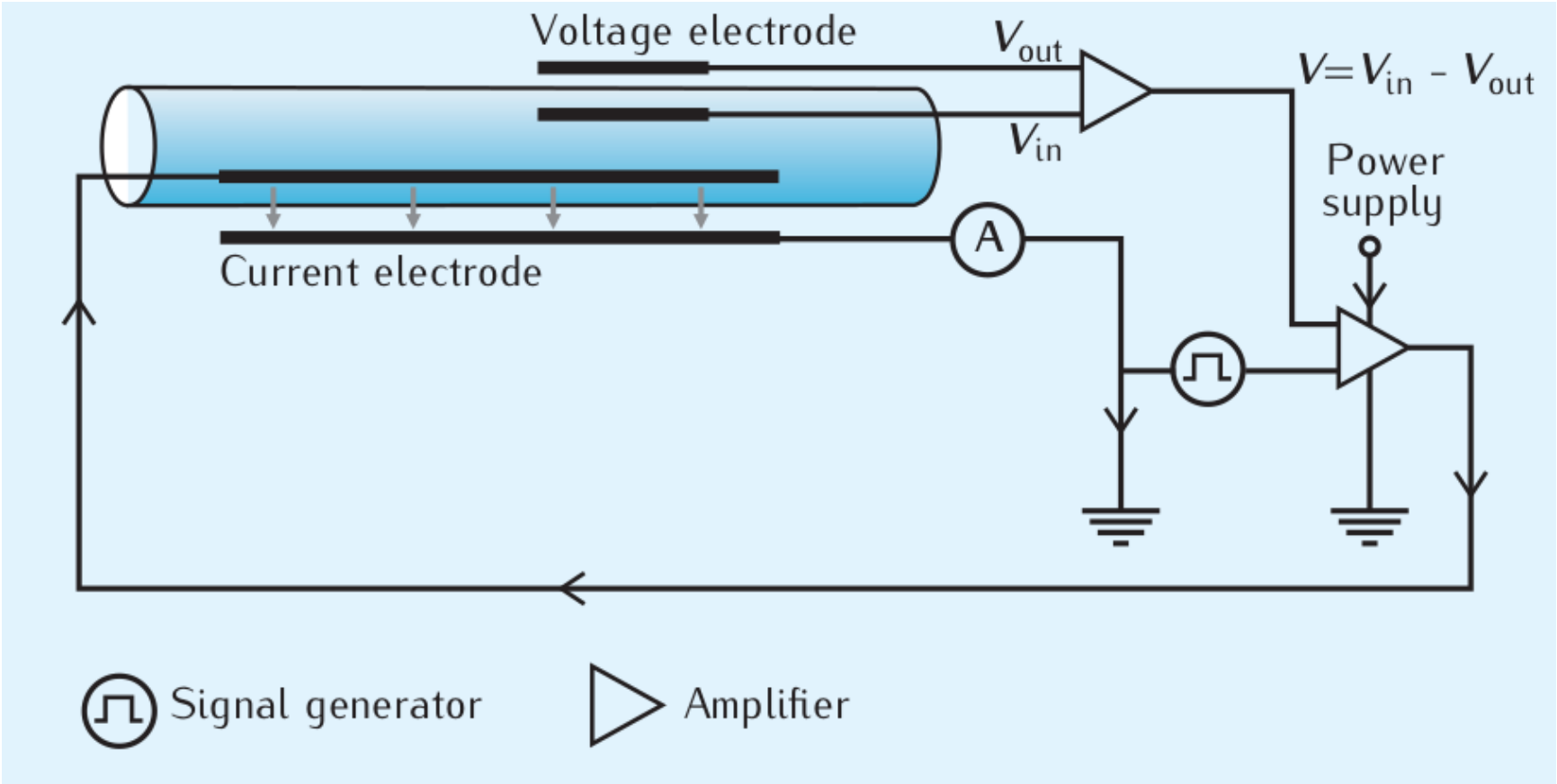
Hodgkin-Huxley: experiments

Made their experiments on the squid giant axon, not the giant squid axon!





Voltage and Space clamp

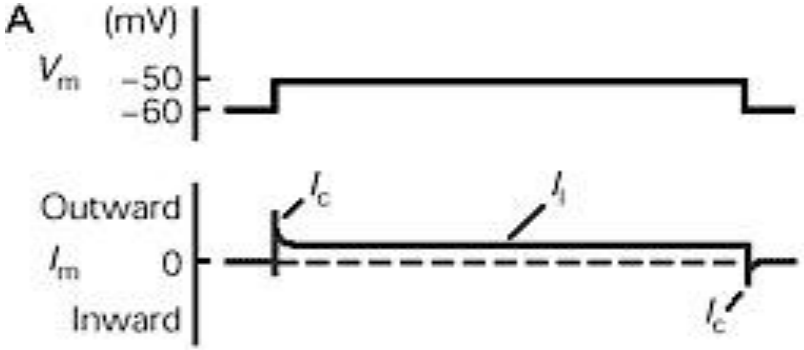


Source: Principles of Computational Modelling in Neuroscience

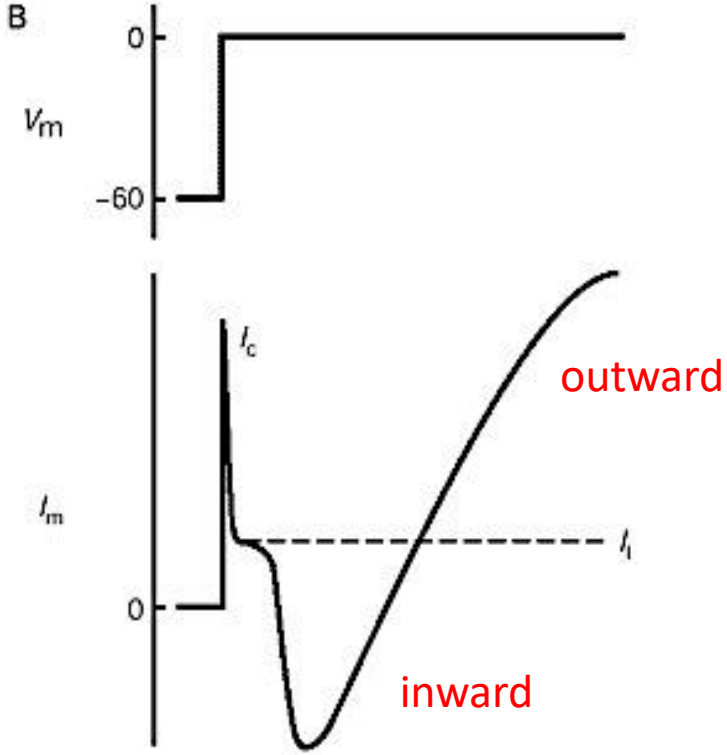
Cole and Marmont in 1940s (Marmont, 1949; Cole, 1968)

http://www.sumanasinc.com/webcontent/animations/content/voltage_clamp.html

The Ionic currents

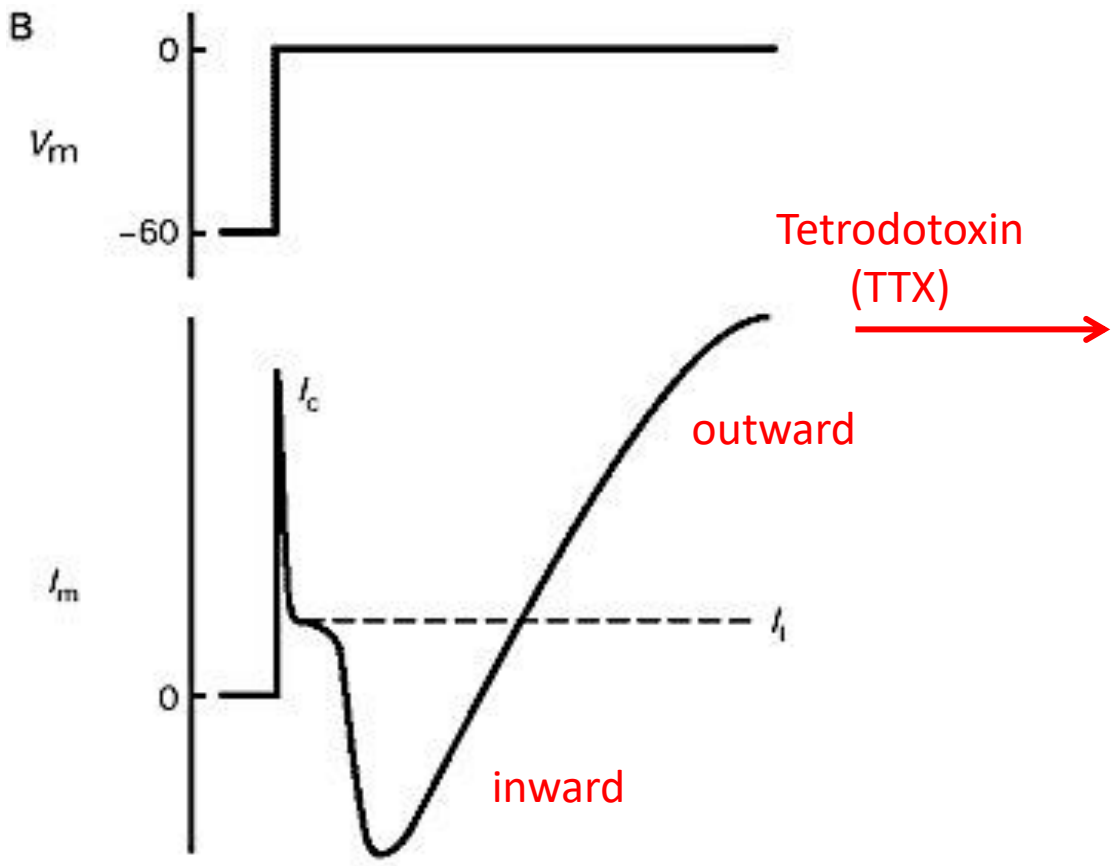


Sub-threshold depolarizing voltage clamp



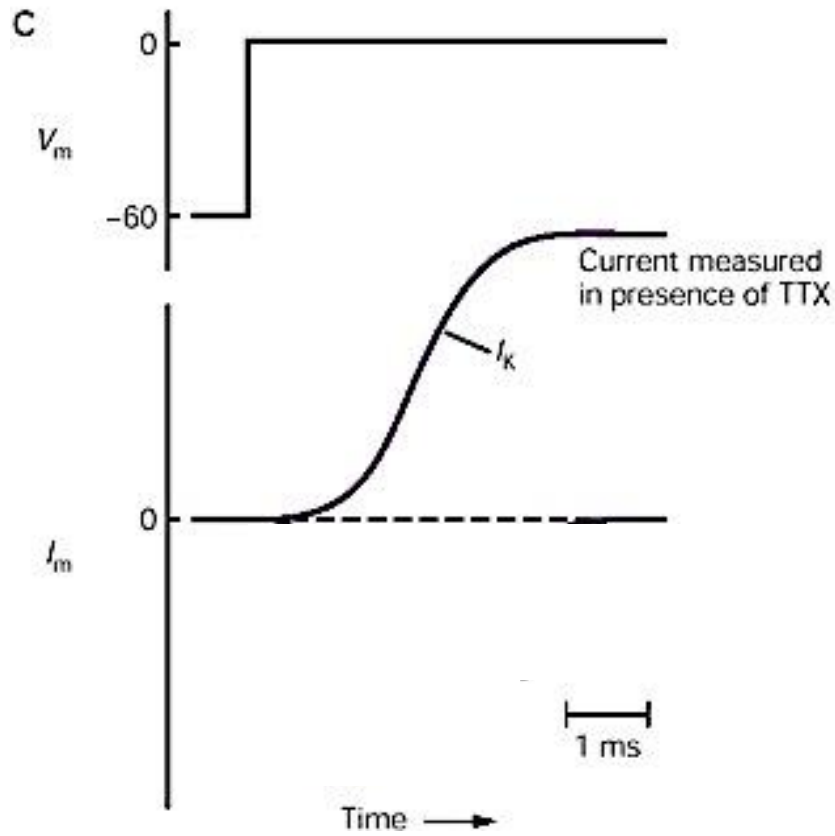
Supra-threshold depolarizing voltage clamp

The Ionic currents



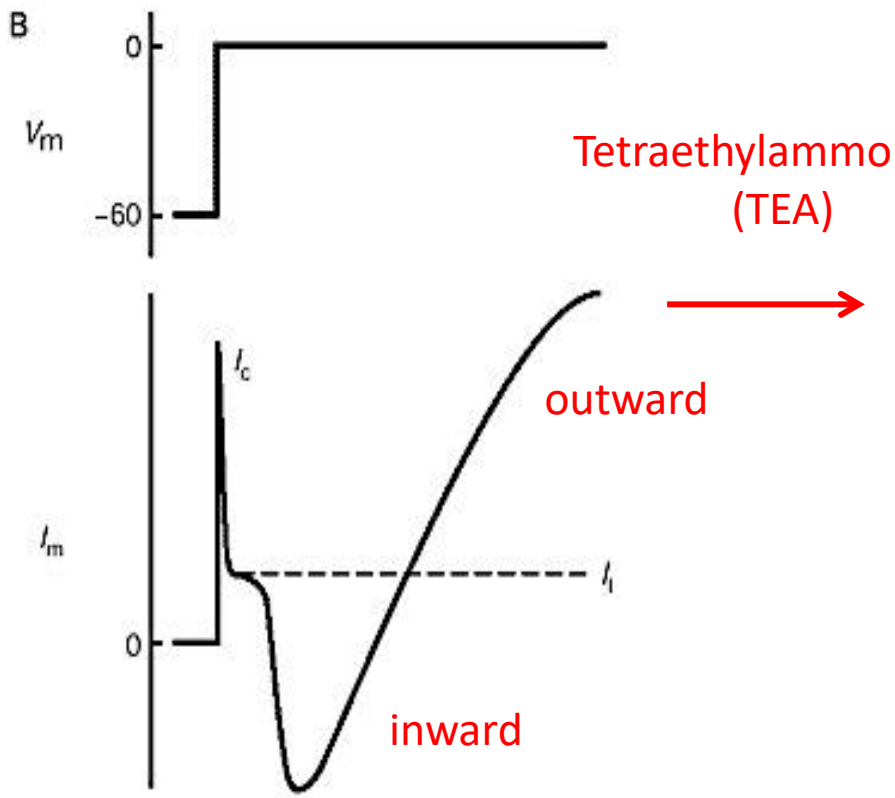
Tetrodotoxin
(TTX)

TTX blocks the early – fast inactivating current

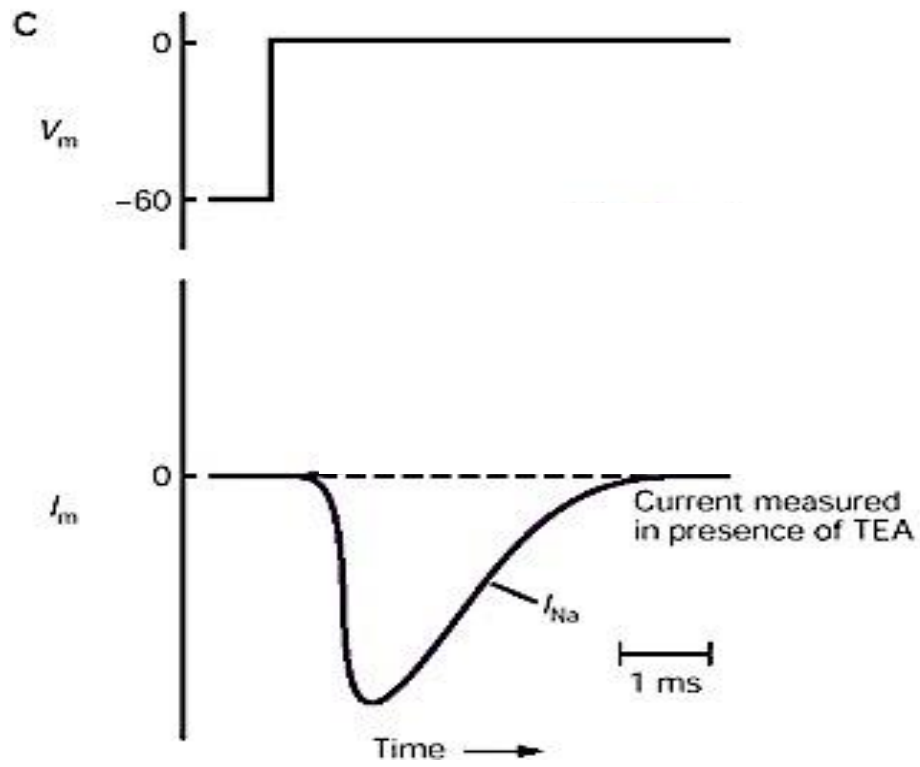


The Ionic currents

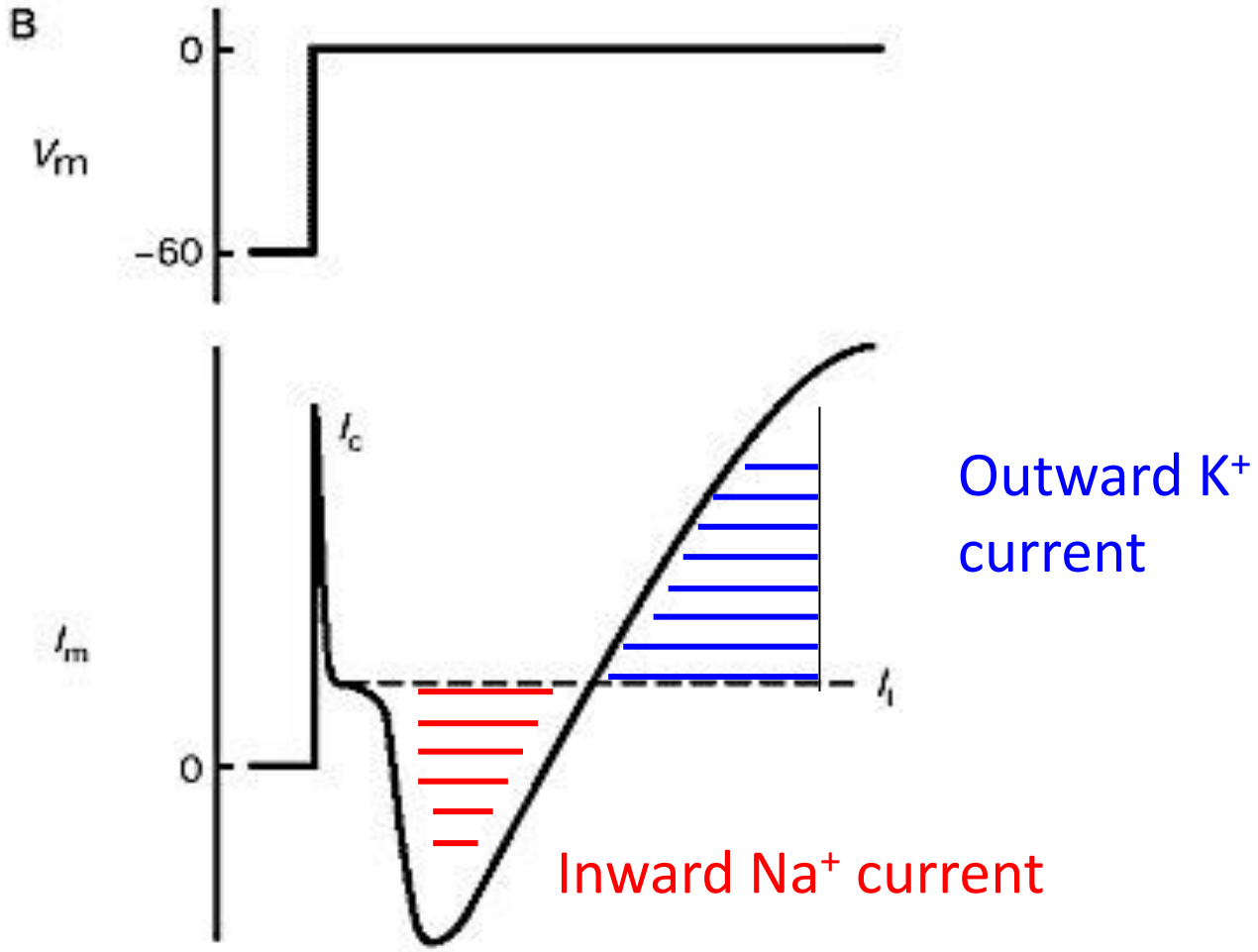
TEA blocks the late – slow non-inactivating current



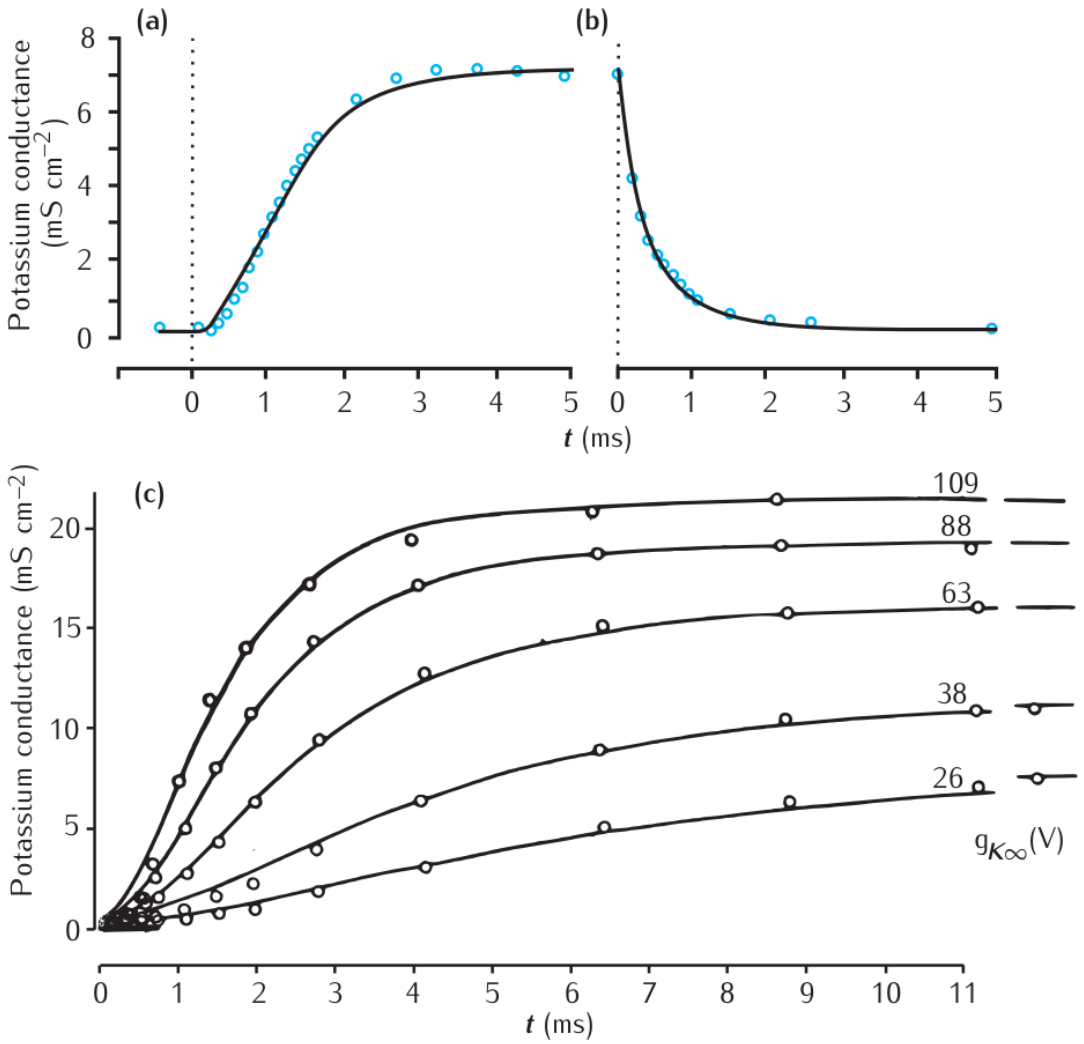
Tetraethylammonium (TEA)



The Ionic currents



The potassium conductance



1. The value that the conductance reaches over time, $\mathbf{g_{K_\infty}}$, approaches a maximum at high holding potentials.

2. The speed at which the limiting conductance is approached becomes faster at higher depolarising holding potentials.

$$g_K = \bar{g}_K n^4$$

The potassium (K^+) current

- Gating particles: either in closed (C) or open position

$$\mathrm{C} \rightleftharpoons \mathrm{O}$$

- The fraction of particles in O state is n
- Rate coefficients: $\boldsymbol{a_n(V)}$ (close to open state)
- Rate coefficients: $\boldsymbol{\beta_{\mathrm{n}}(V)}$ (open to close state)

rate law or 1st order kinetic equation

$$\frac{dn}{dt} = a_n(1-n) - \beta_n n$$

The potassium (K^+) current

$$n(t) = n_\infty(V) - (n_\infty(V) - n_0)\exp\left(-\frac{t}{\tau_n(V)}\right)$$

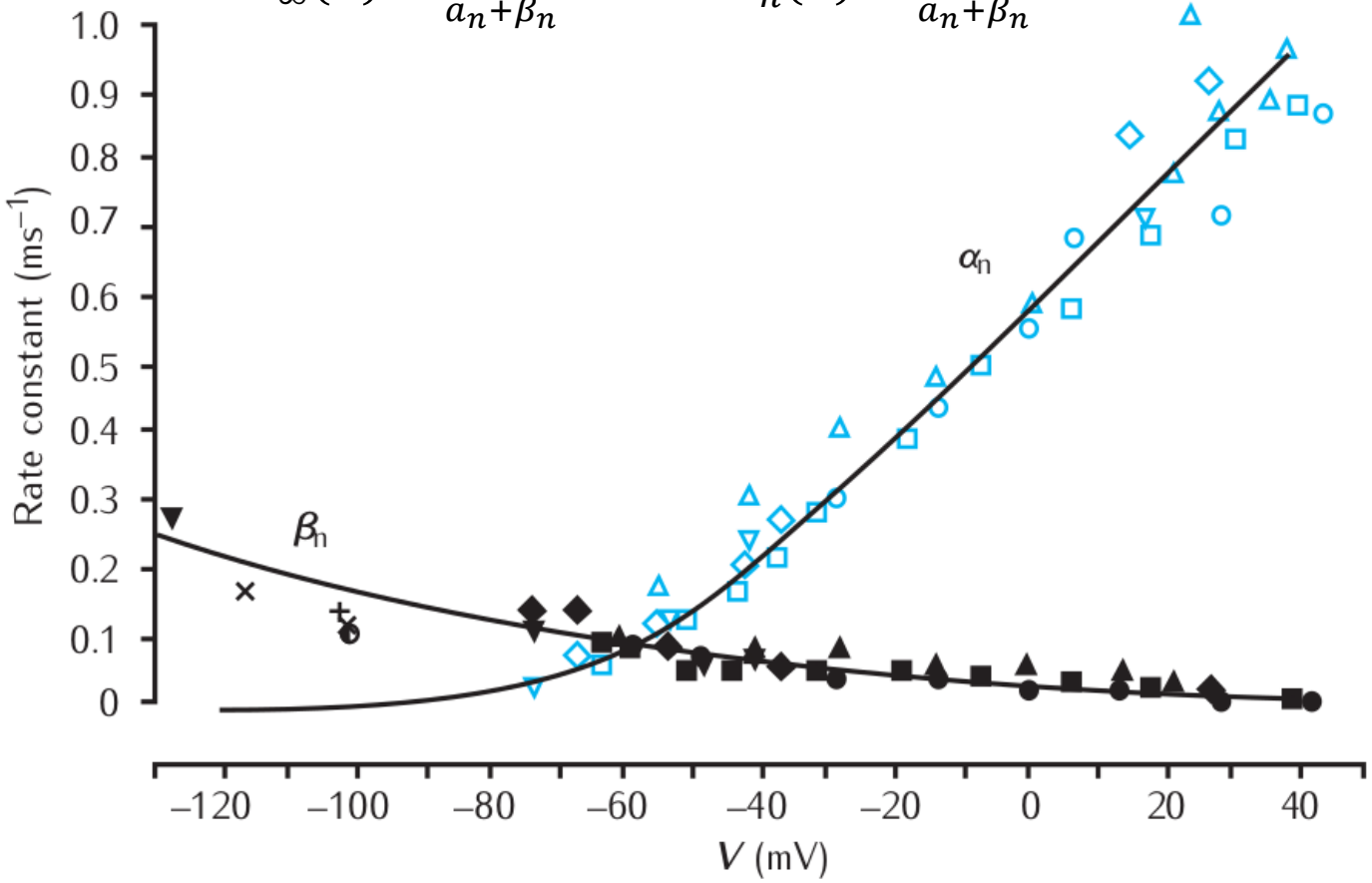
where n_0 is the value of n at the start of the step, time zero

The variables $n_\infty(V)$ and $\tau_n(V)$ are related to rate coefficients

$$n_\infty(V) = \frac{a_n}{a_n+\beta_n} \quad \text{and} \quad \tau_n(V) = \frac{1}{a_n+\beta_n}$$

Rate coefficients

$$n_\infty(V) = \frac{a_n}{a_n+\beta_n} \quad \text{and} \quad \tau_n(V) = \frac{1}{a_n+\beta_n}$$



Potassium current: complete story

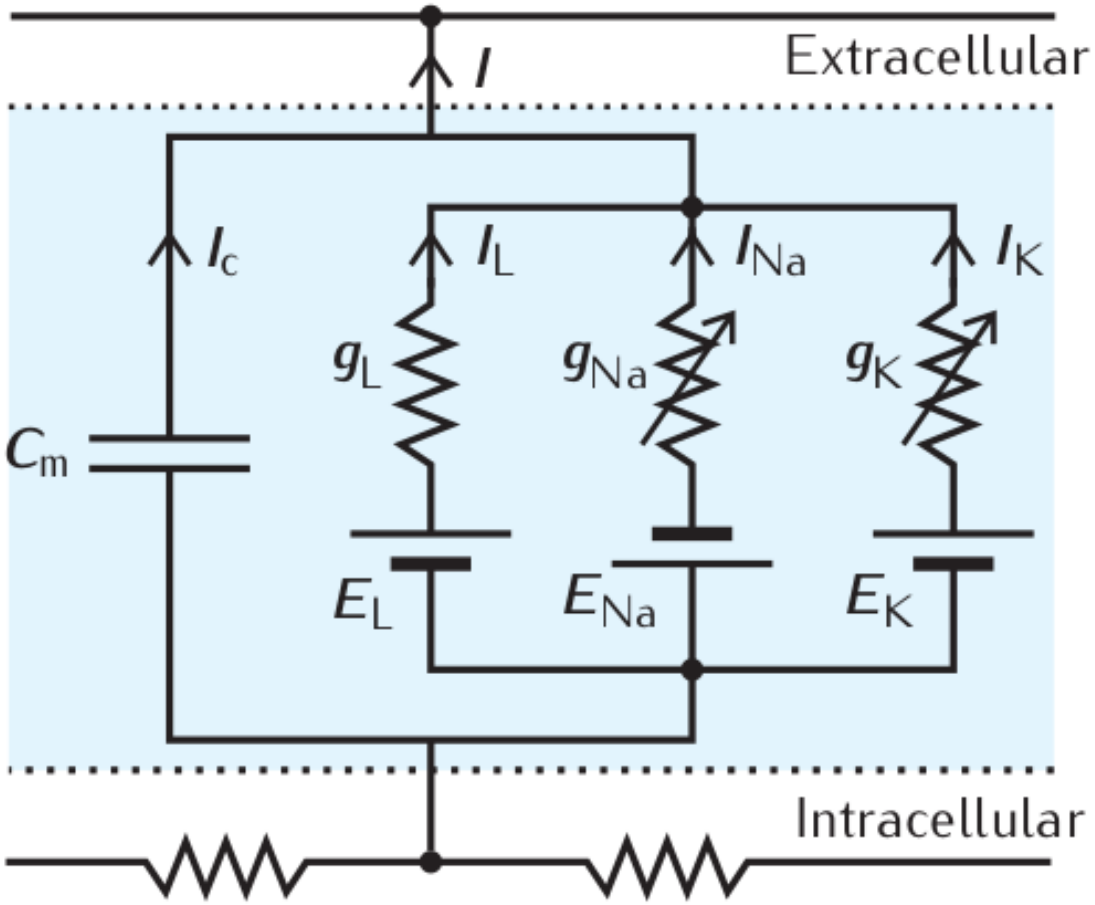
$$I_K = \bar{g}_K n^4 (V - E_K)$$

$$\frac{dn}{dt} = a_n(1 - n) - \beta_n n$$

$$a_n = 0.01 \cdot \frac{V + 55}{1 - \exp(-(V + 55)/10)}$$

$$\beta_n = 0.125 \cdot \exp(-(V + 65)/80)$$

HH equivalent electrical circuit



Conductances in cell membrane varied with time and voltage

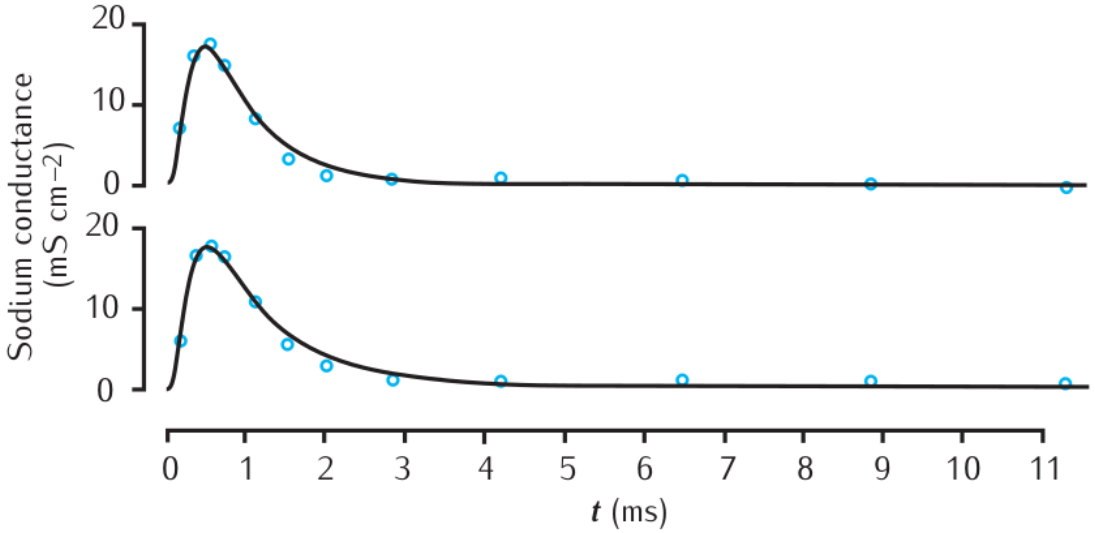
$$g_{Na} = g_{Na}(V)$$

$$g_K = g_K(V)$$

$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - g_{Na}(V - E_{Na}) - g_K(V - E_K) + I$$

The sodium current

- **Activation** and **inactivation** phase
- Two gating variables, h and m



Inactivation gating

- Possible states:
  - Not inactivated
  - Inactivated
- First order kinetic equation (similar to n)

$$\frac{dh}{dt} = a_h(1 - h) - \beta_h h$$

- Rate coefficients: $a_h(V), \beta_h(V)$
- Parameters: $h_\infty(V), h_0(V), \tau_h(V)$

Open/Close gating

- Possible states:
    - Open
    - Close
- First order kinetic equation (similar to n, h)

$$\frac{dm}{dt} = a_m(1 - m) - \beta_m m$$

- Rate coefficients: $a_m(V), \beta_m(V)$
- Parameters: $m_\infty(V), m_0(V), \tau_m(V)$

Sodium current: complete story

$$I_{\mathrm{Na}} = \bar{g}_{\mathrm{Na}} m^3 h (V - E_{\mathrm{Na}})$$

Open-close gating variable

$$\frac{dm}{dt} = a_m(1 - m) - \beta_m m$$

$$a_m = 0.1 \frac{V + 40}{1 - \exp(-(V + 40)/10)}$$

$$\beta_m = 4\exp(-(V + 65)/18)$$

Inactivating gating variable

$$\frac{dh}{dt} = a_h(1 - h) - \beta_h h$$

$$a_h = 0.07\exp(-(V + 65)/20)$$

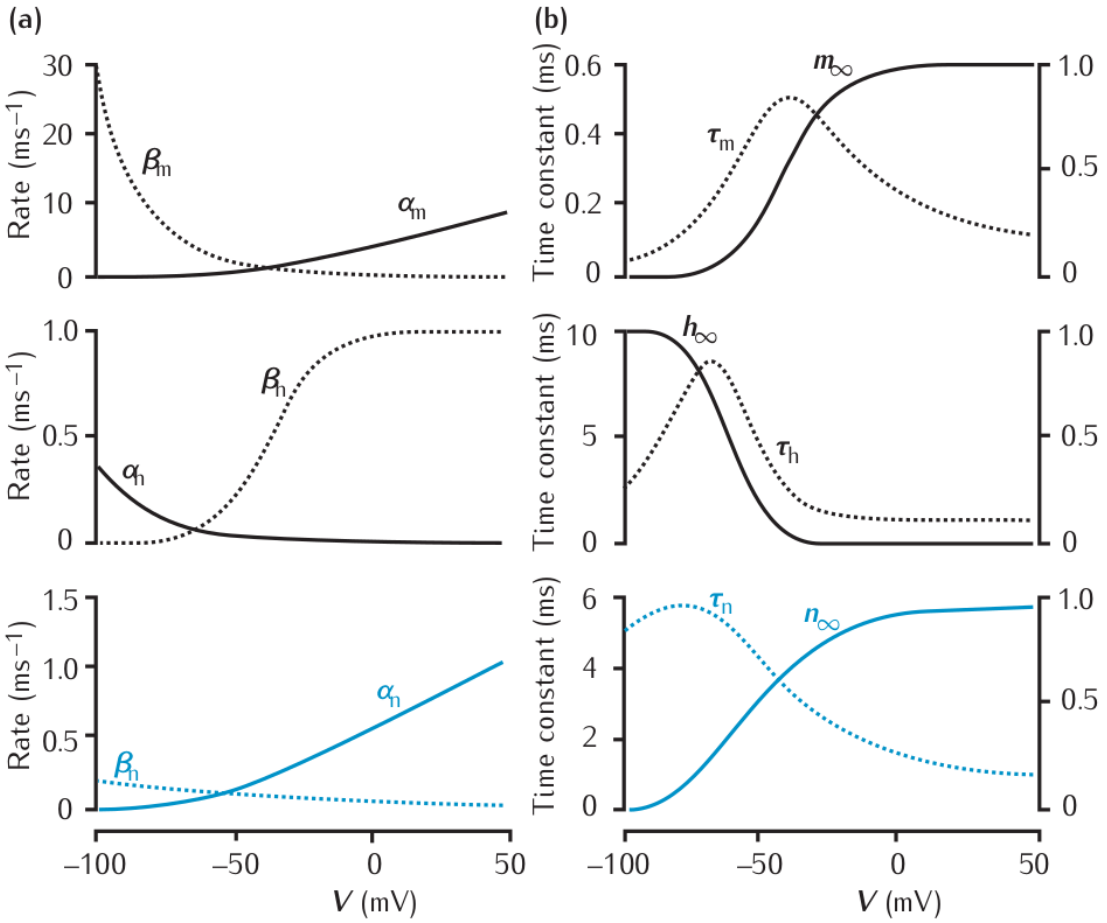
$$\beta_h = \frac{1}{1 + \exp(-(V + 35)/10)}$$

HH: The complete model

$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 h(V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

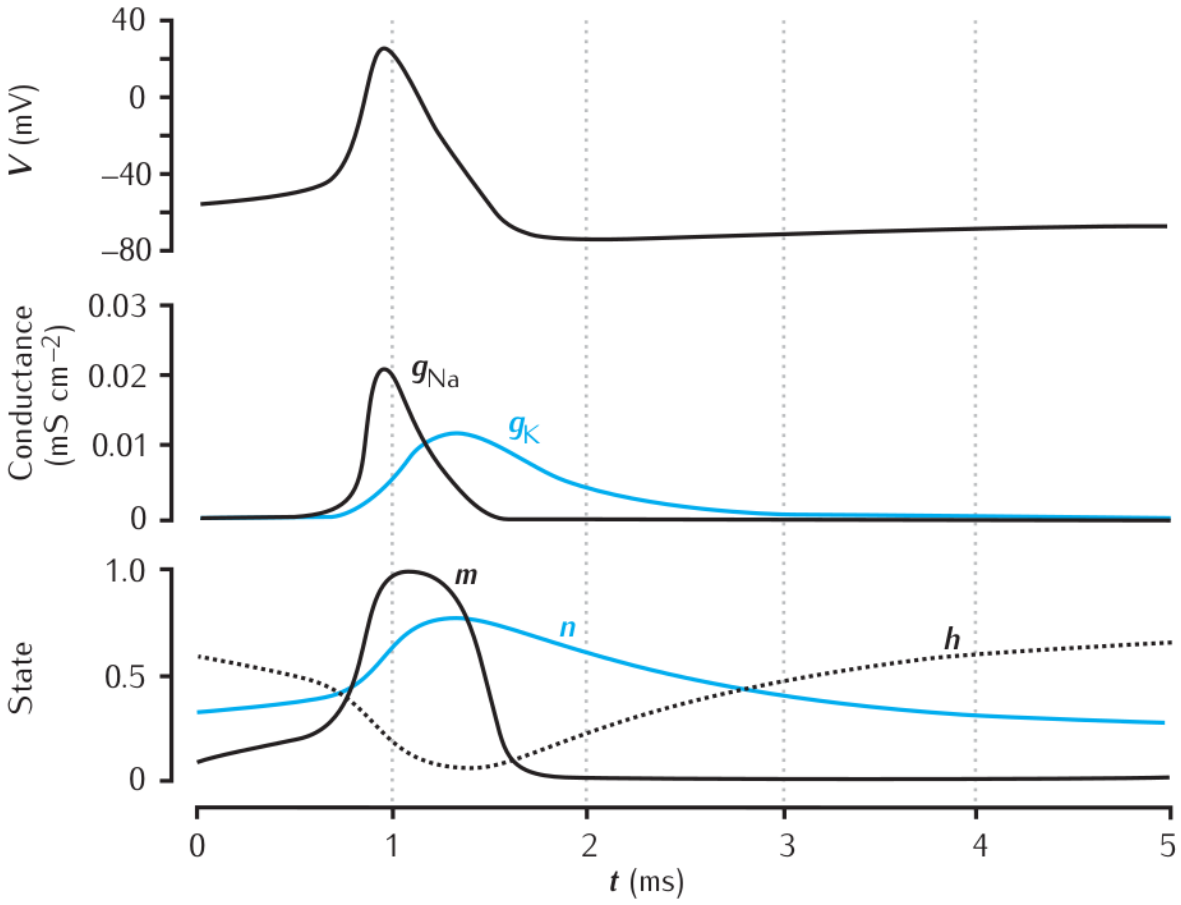
$$n_\infty(V) = \frac{a_n}{a_n + \beta_n}$$

$$\tau_n(V) = \frac{1}{a_n + \beta_n}$$



HH: The complete model

$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 h (V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$



HH model is too complex: four differential equations!

I&F models are too simple!

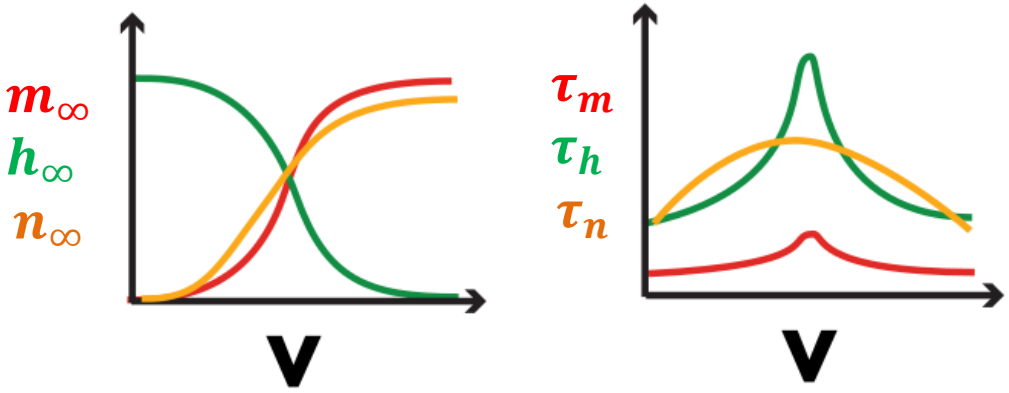
HH model: Reduction

$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m_\infty^3 h(V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

$$\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}$$

$$\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}$$

$$\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}$$



This is a four dimensional system, but …

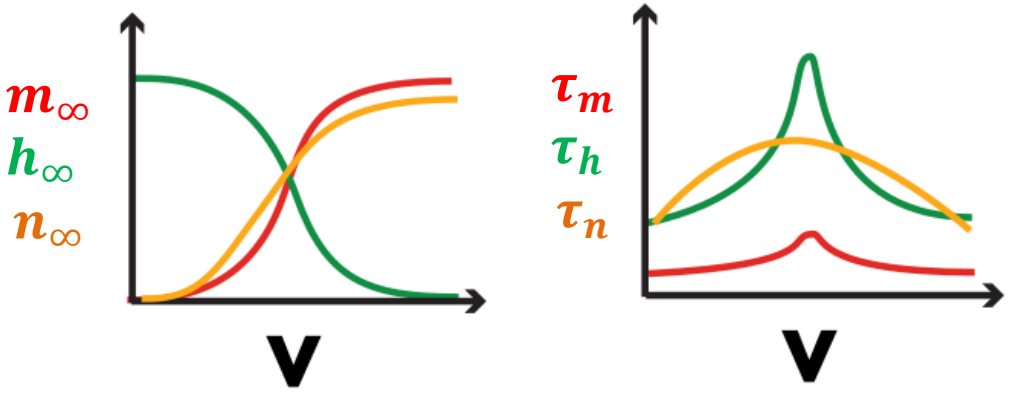
HH model: Reduction

$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 h (V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

$$\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}$$

$$\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}$$

$$\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}$$



This is a four dimensional system, but …

m is almost always m_∞

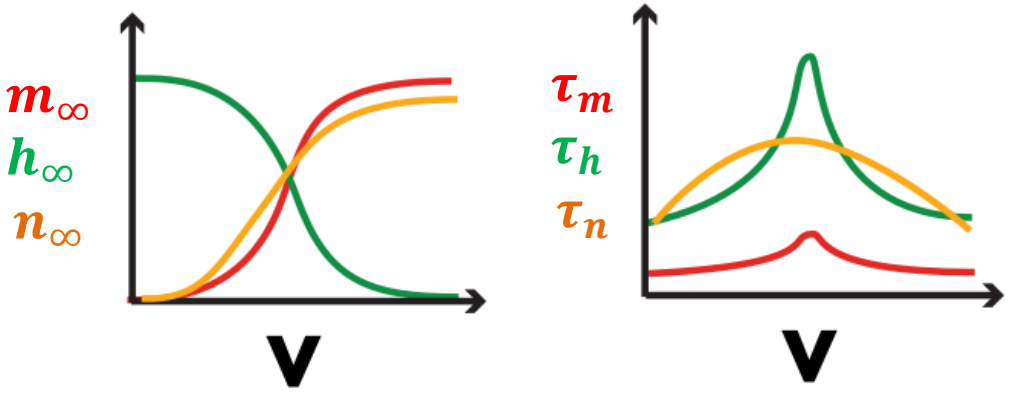
HH model: Reduction

$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 h (V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

$$\cancel{\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}}$$

$$\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}$$

$$\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}$$



This is a four dimensional system, but …

m is almost always m_∞

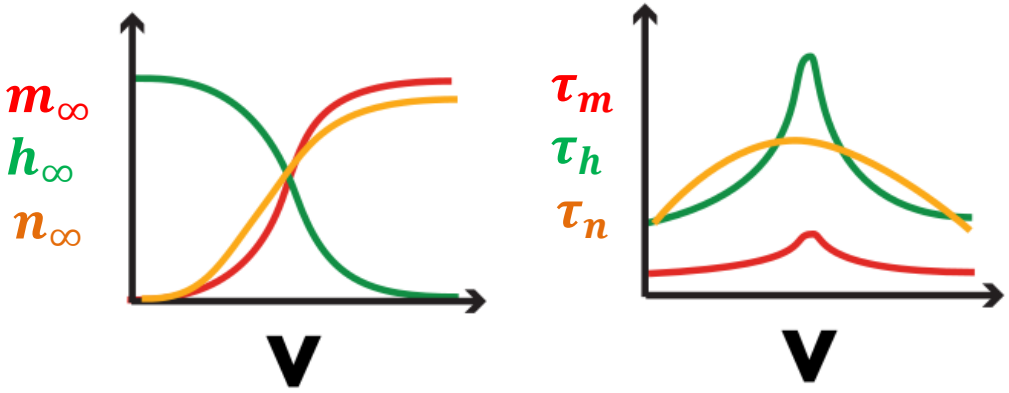
HH model: Reduction

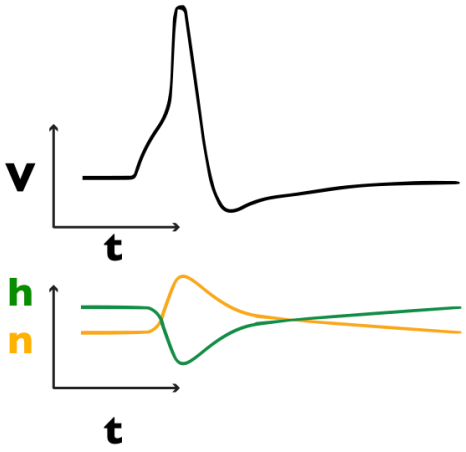
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 h (V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

$$\cancel{\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}}$$

$$\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}$$

$$\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}$$





This is a four dimensional system, but …

m is almost always m_∞
h and n are mirroring each other

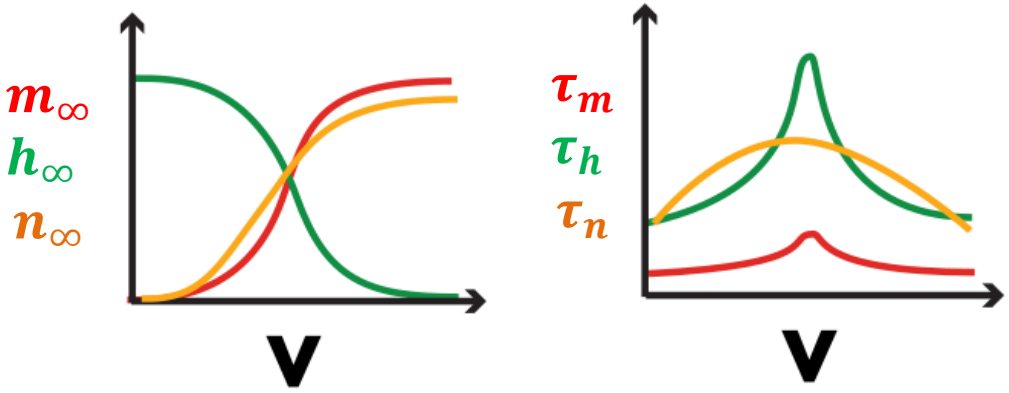
HH model: Reduction

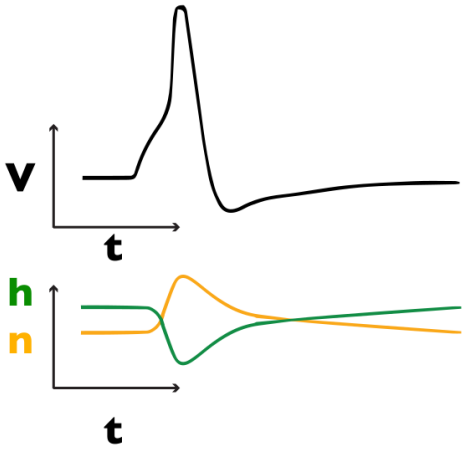
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 h (V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

$$\cancel{\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}}$$

$$\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}$$

$$\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}$$





This is a four dimensional system, but …

m is almost always m_∞

h and n are mirroring each other

$$an = 1 - h$$

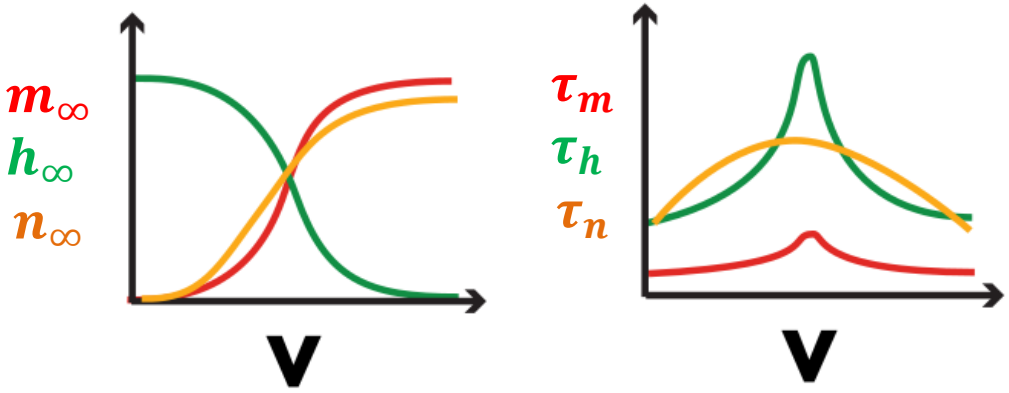
HH model: Reduction

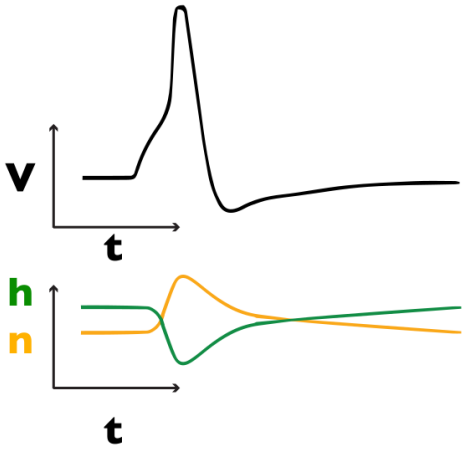
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 (1 - w)(V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

$$\cancel{\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}}$$

$$\cancel{\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}}$$

$$\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}$$





This is a four dimensional system, but …

m is almost always m_∞

h and n are mirroring each other

$$an = 1 - h = w$$

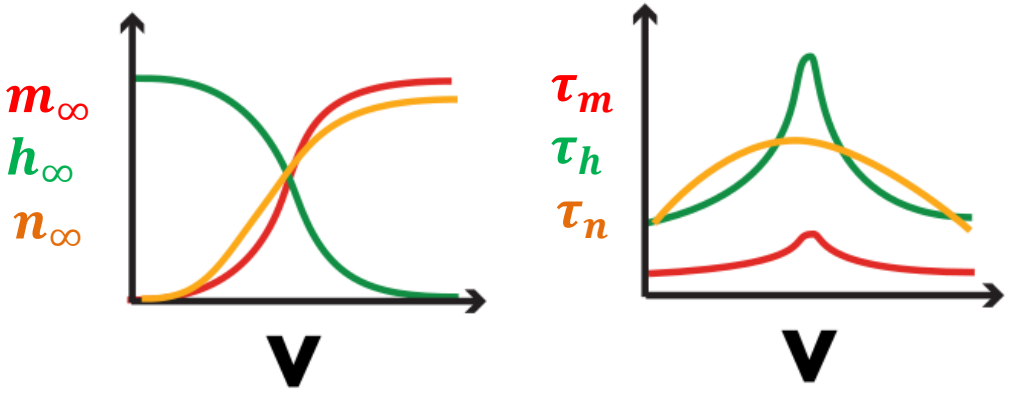
HH model: Reduction

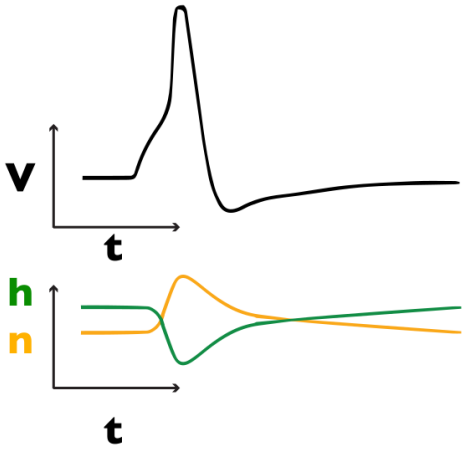
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 (1 - w)(V - E_{Na}) - \bar{g}_K \left(\frac{w}{a}\right)^4 (V - E_K) + I$$

$$\cancel{\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}}$$

$$\cancel{\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}}$$

$$\cancel{\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}}$$





This is a four dimensional system, but …

m is almost always m_∞

h and n are mirroring each other

$$an = 1 - h = w$$

HH model: Reduction

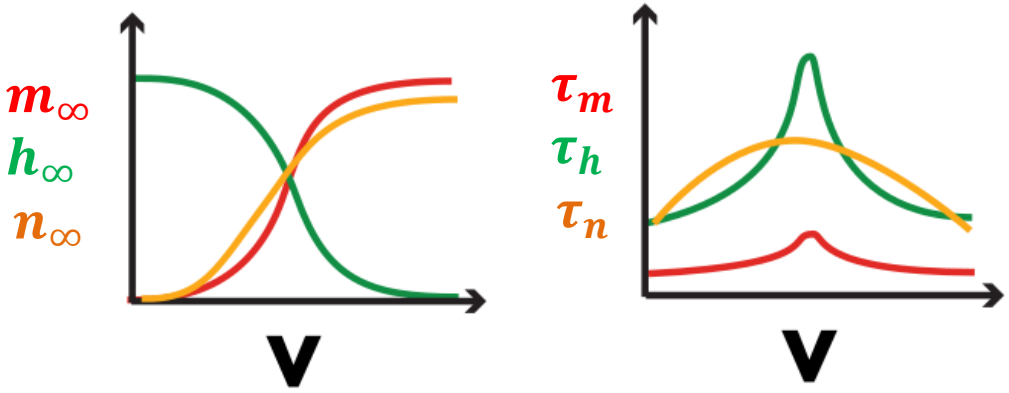
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 (1 - w)(V - E_{Na}) - \bar{g}_K \left(\frac{w}{a}\right)^4 (V - E_K) + I$$

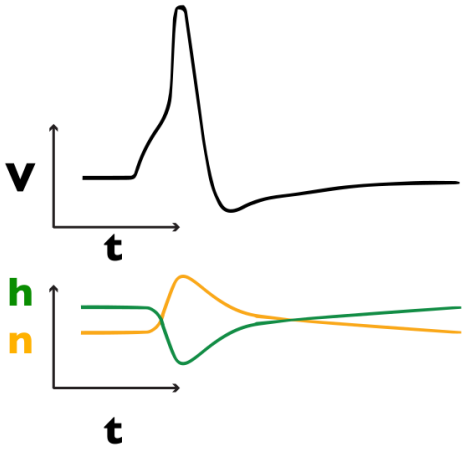
$$\cancel{\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m}}$$

$$\cancel{\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h}}$$

$$\cancel{\frac{dn}{dt} = \frac{n_\infty - n}{\tau_n}}$$

$$\frac{dw}{dt} = \frac{w_\infty - w}{\tau_n}$$





This is a four dimensional system, but …

m is almost always m_∞

h and n are mirroring each other

$$an = 1 - h = w$$

HH model: Reduction

$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na} m^3 h(V - E_{Na}) - \bar{g}_K n^4 (V - E_K) + I$$

$$\tau \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = F\big(V(t), w(t)\big)$$

$$\frac{d\boldsymbol{w}}{dt} = \frac{\boldsymbol{w}_\infty - \boldsymbol{w}}{\boldsymbol{\tau_n}}$$

$$\tau_n \frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = G\big(w(t), V(t)\big)$$

HH model: Reduction

$$\tau \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = F\big(V(t), w(t)\big)$$

$$\tau_n \frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = G\big(w(t), V(t)\big)$$

HH model: Reduction

$$\tau \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = F\big(V(t), w(t)\big) \Rightarrow \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = V - \frac{1}{3}V^3 - w$$

$$\tau_n \frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = G\big(w(t), V(t)\big) \Rightarrow \frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = \mathrm{a} + \mathrm{b}V - w$$

HH model: Reduction

$$\tau \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = F\big(V(t), w(t)\big) \Rightarrow \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = V - \frac{1}{3}V^3 - w$$

$$\tau_n \frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = G\big(w(t), V(t)\big) \Rightarrow \frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = a + bV - w$$

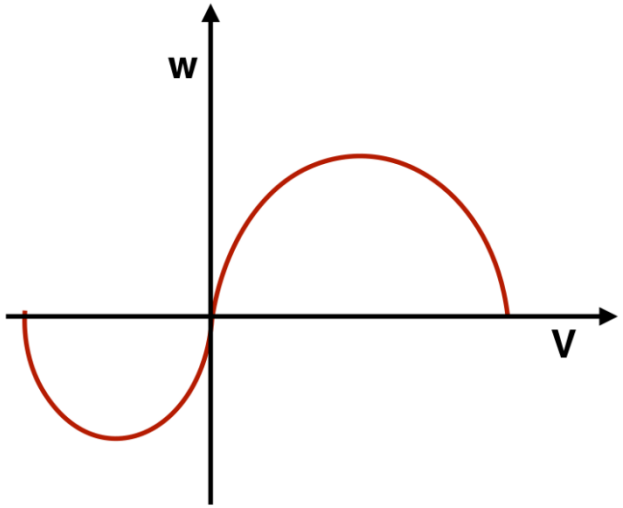
FitzHugh-Nagumo Model

FitzHugh-Nagumo model

$$\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = V - \frac{1}{3}V^3 - w$$

$$\frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = \mathrm{a} + \mathrm{b}V - w$$

$$\frac{dV}{dt} = 0 \Leftrightarrow V - \frac{1}{3}V^3 - w = 0$$



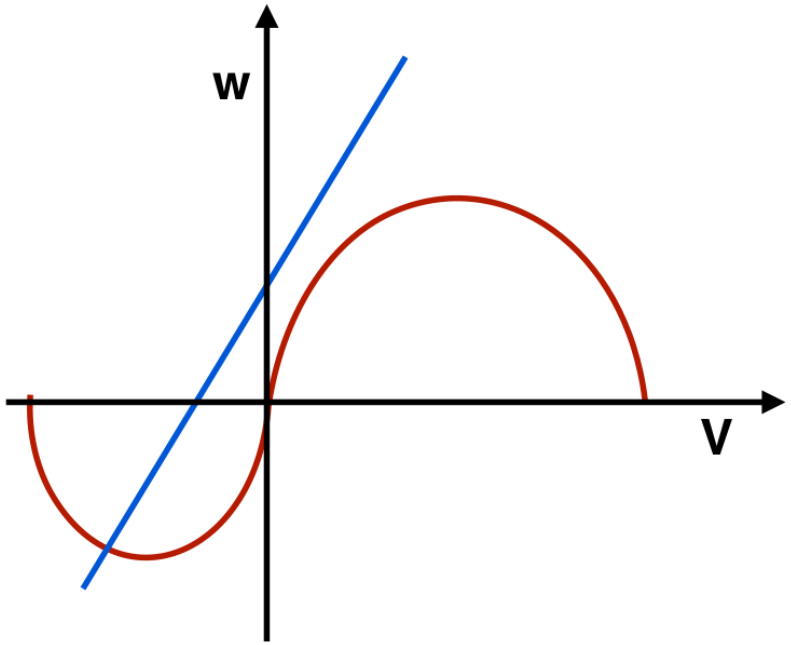
FitzHugh-Nagumo model

$$\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = V - \frac{1}{3}V^3 - w$$

$$\frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = \mathrm{a} + \mathrm{b}V - w$$

$$\frac{dV}{dt} = 0 \Leftrightarrow V - \frac{1}{3}V^3 - w = 0$$

$$\frac{dw}{dt} = 0 \Leftrightarrow \mathrm{a} + \mathrm{b}V - w = 0$$



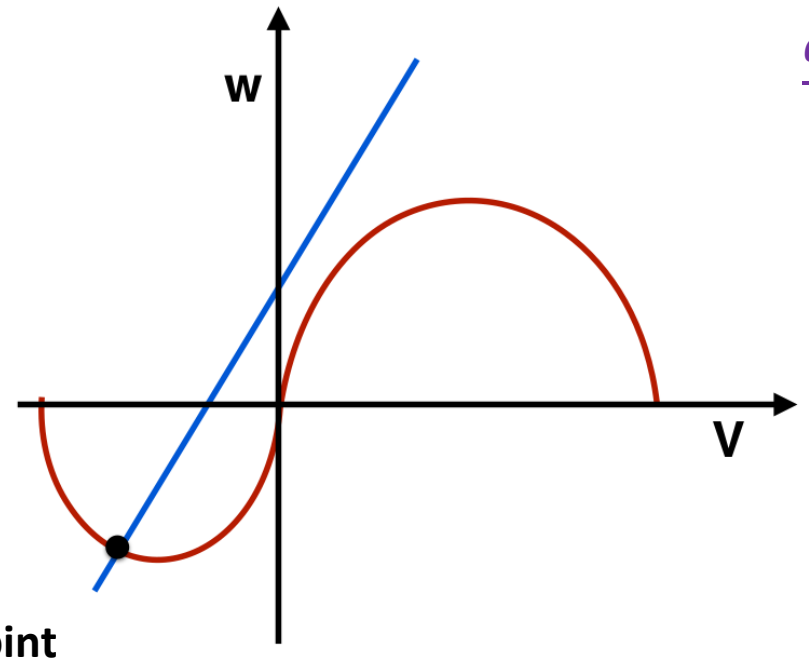
FitzHugh-Nagumo model

$$\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = V - \frac{1}{3}V^3 - w$$

$$\frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = \mathrm{a} + \mathrm{b}V - w$$

$$\frac{d\boldsymbol{V}}{d\boldsymbol{t}} = \mathbf{0} \Leftrightarrow \boldsymbol{V} - \frac{\mathbf{1}}{\mathbf{3}}\boldsymbol{V}^{\mathbf{3}} - \boldsymbol{w} = \mathbf{0}$$

$$\frac{d\boldsymbol{w}}{d\boldsymbol{t}} = \mathbf{0} \Leftrightarrow \mathbf{a} + \boldsymbol{bV} - \boldsymbol{w} = \mathbf{0}$$



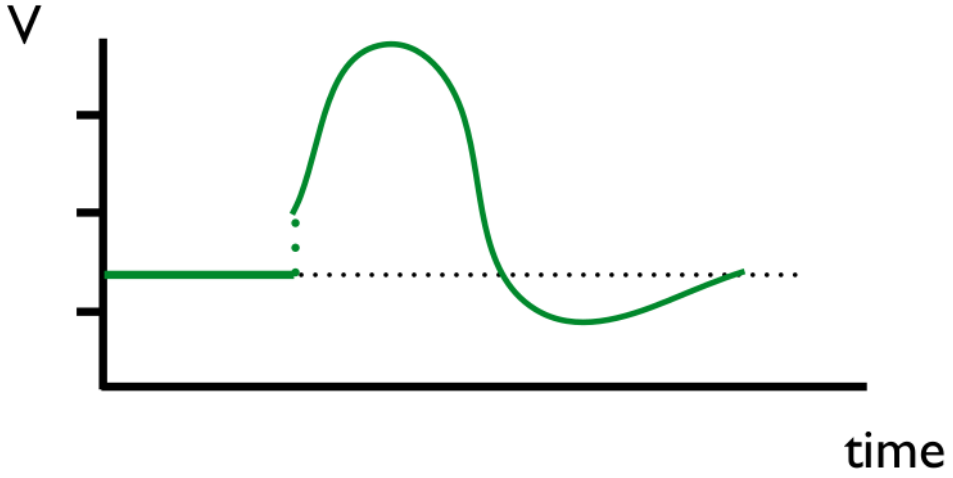
(phase plane analysis)

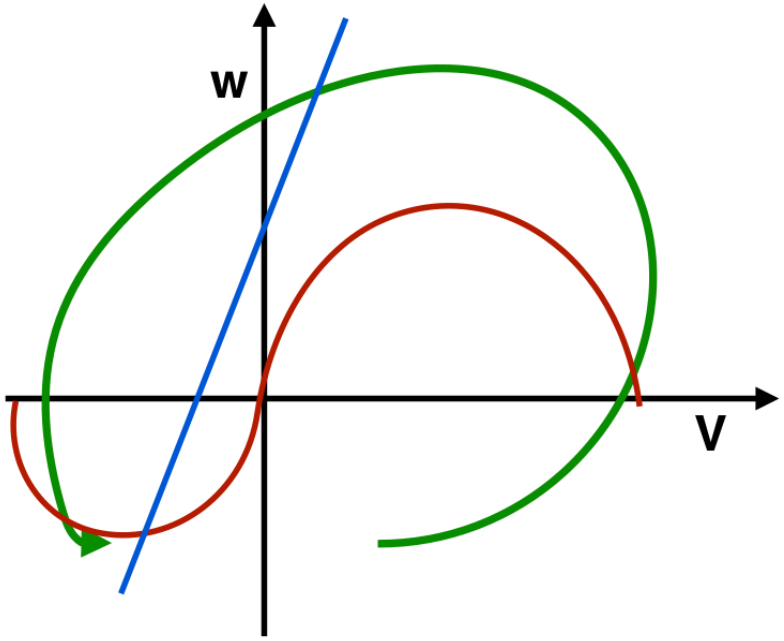
Fixed point

FitzHugh-Nagumo model

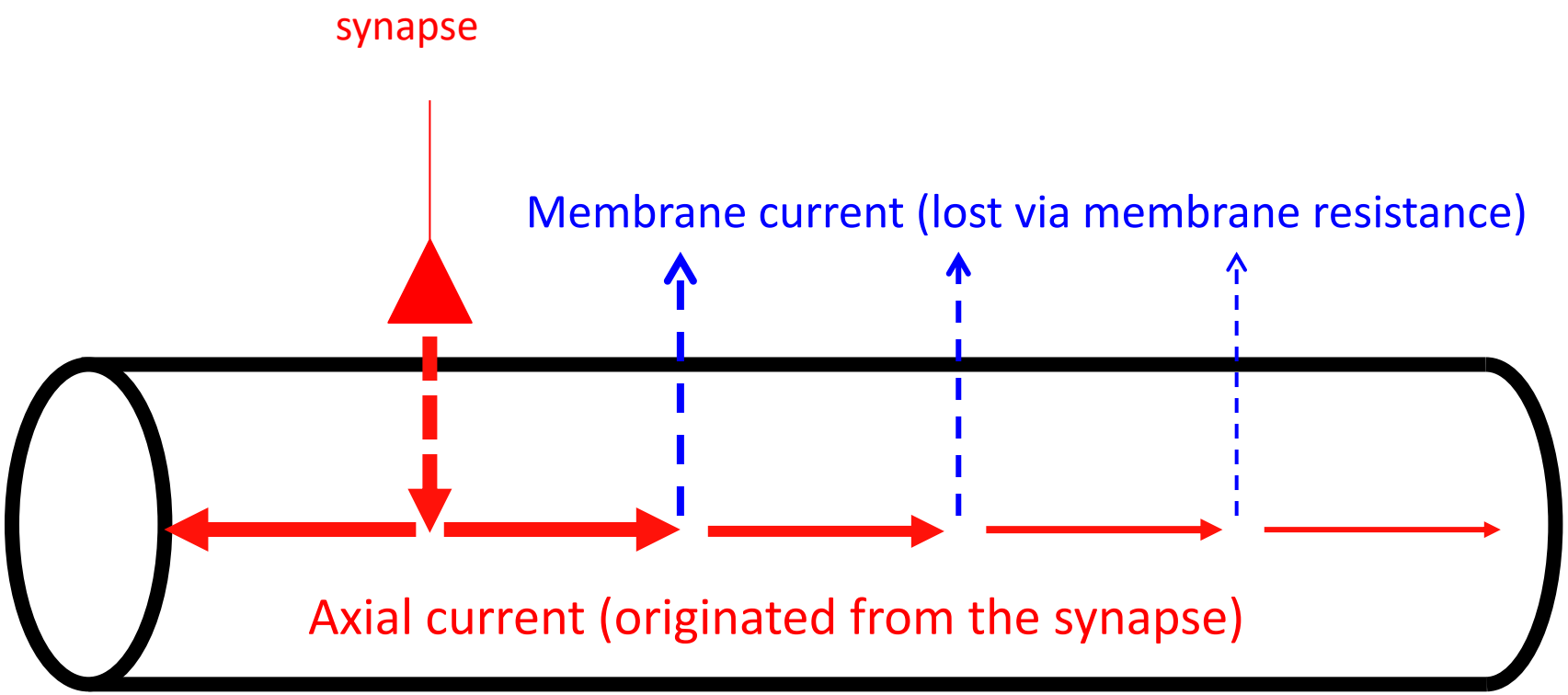
$$\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}\boldsymbol{t}} = V - \frac{1}{3}V^3 - w$$

$$\frac{\mathrm{d}\boldsymbol{w}}{\mathrm{d}\boldsymbol{t}} = \mathrm{a} + \mathrm{b}V - w$$





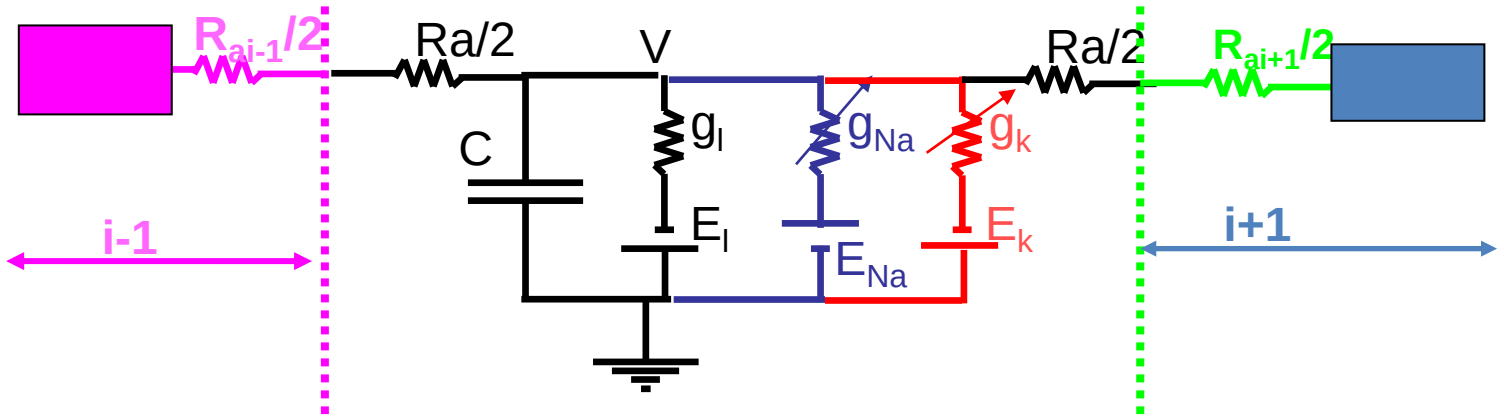
Cable Theory and Compartmental Modeling



$$C_m \frac{\partial V}{\partial t} = -\sum_k I_{i,k}(x) + \frac{d}{4R_a}\frac{\partial^2 V}{\partial x^2} + \frac{I_e(x)}{\pi d},$$

Rall, 1964

Cable Theory and Compartmental Modeling

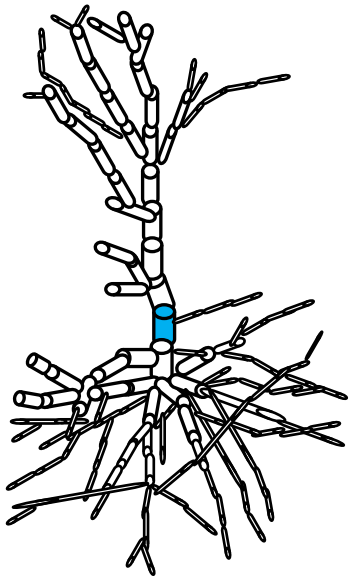


$$C_m \frac{dV}{dt} = -g_l(V - E_l) - g_{\mathrm{Na}}(V - E_{\mathrm{Na}}) - g_{\mathrm{K}}(V - E_{\mathrm{K}}) + 2 \cdot \frac{V_{i+1} - V_i}{R_{a,i+1} + R_{a,i}} + 2 \cdot \frac{V_{i-1} - V_i}{R_{a,i-1} + R_{a,i}} + I_{inj}$$

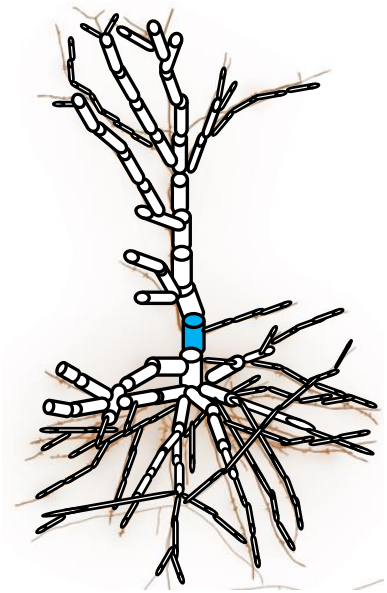
$$C_{\mathrm{m}} \frac{\partial V}{\partial t} = -\sum_k I_{\mathrm{i},k}(x) + \frac{d}{4R_{\mathrm{a}}} \frac{\partial^2 V}{\partial x^2} + \frac{I_{\mathrm{e}}(x)}{\pi d},$$

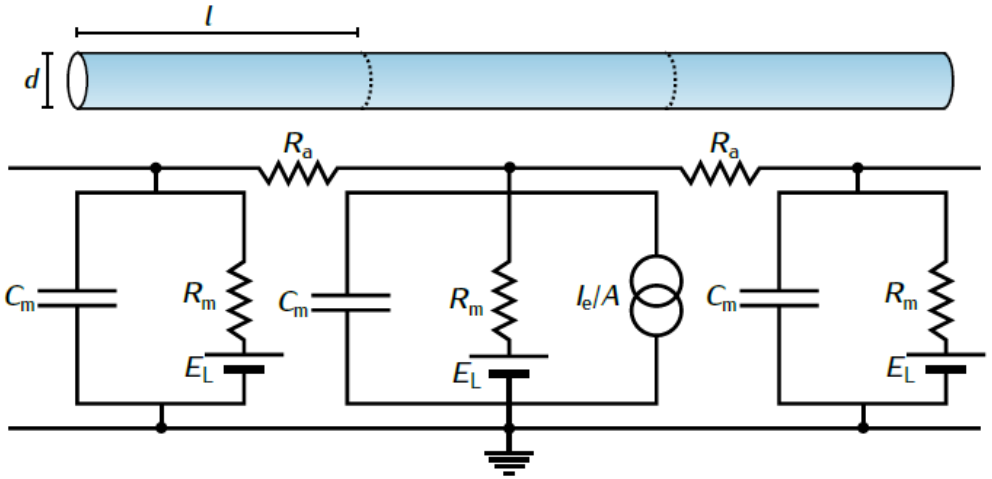
Cable Theory and Compartmental Modeling

Cable Theory and Compartmental Modeling

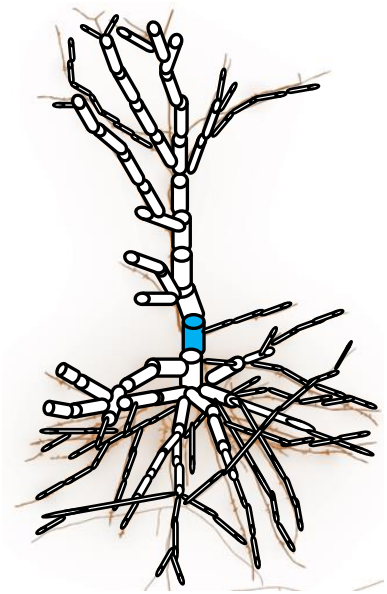


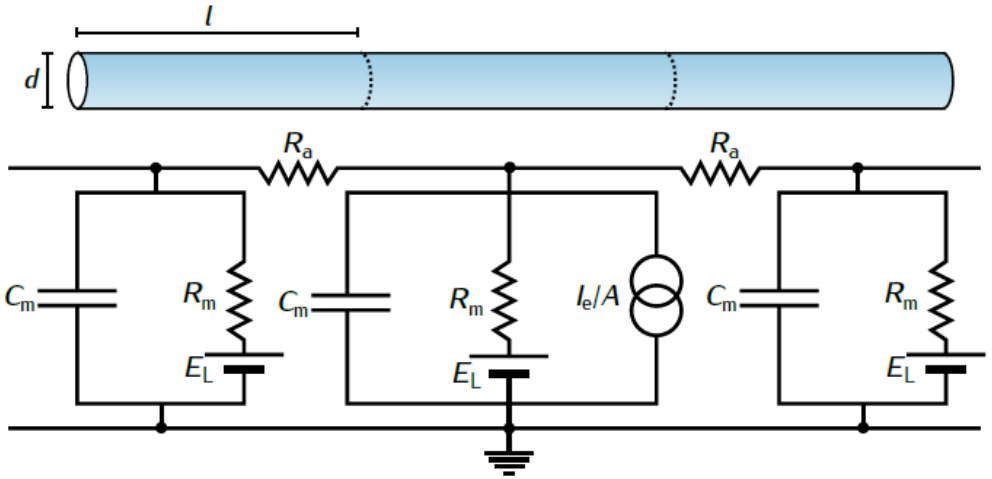
Cable Theory and Compartmental Modeling





Cable Theory and Compartmental Modeling





Space constant (λ)

$$\lambda = \sqrt{\frac{R_m d}{4R_a}}$$

Table 2.3 Passive quantities

Quantity	Description	Typical units	Relationships
d	Diameter of neurite	µm	
l	Length of compartment	µm	
R_m	Specific membrane resistance	$\Omega\,cm^2$	
C_m	Specific membrane capacitance	$\mu F\,cm^{-2}$	
R_a	Specific axial resistance (resistivity)	$\Omega\,cm$	
r_m	Membrane resistance per inverse unit length	$\Omega\,cm$	$r_m = \frac{R_m}{\pi d}$
c_m	Membrane capacitance per unit length	$\mu F\,cm^{-1}$	$c_m = C_m \pi d$
r_a	Axial resistance per unit length	Ω/cm	$r_a = \frac{4R_a}{\pi d^2}$
V	Membrane potential	mV	
E_m	Leakage reversal potential due to different ions	mV	
I	Membrane current density	$\mu A\,cm^{-2}$	
I_e	Injected current	nA	
I_c	Capacitive current	nA	
I_i	Ionic current	nA	

Next Practical

- Dendritic Computations and NEURON simulation environment!